Airfoil Co-Kriging

Load the NACA airfoil coordinates and predict. The coordinates should be arranged in one sense. Meaning, that `XB` and `YB` are either in a clockwise or anti-clockwise arrangement.

Exact solution

The true velocity is computed using the BEM code. GRID discretization used:

```
{python}
# Grid parameters
nGridX   = 30
# X-grid for streamlines and contours
nGridY   = 30
# Y-grid for streamlines and contours
xVals    = [min(XB)-0.5, max(XB)+0.5]
# X-grid extents [min, max]
yVals    = [min(YB)-0.2, max(YB)+0.2]
# Y-grid extents [min, max]
```

In [66]:
```
# imports
import numpy as np
import matplotlib.pyplot as plt
from shapely.geometry import Polygon, Point
# from scipy.interpolate import griddata
from matrices import CovM, CovFn, Umat
from covderivatives2D import AnalyticalSqExpdiff2D
from calibration import LOOMSECK2D, LOOVARCK2D, adamopt, Adam
import matplotlib.cm as cm
from matplotlib.colors import Normalize
```

In [67]:
```
# Image path to save the images.
IMGPATH = r'IMAGES/'

# load the NACA airfoil data
NACAname = r'2424'
XB, YB, nX, nY = np.loadtxt('data/NACA'+NACAname+'UIUCnormalvecs.dat',
                            unpack=True)
XX, YY, Vx, Vy = np.loadtxt('data/NACA'+NACAname+'BEMflow.dat',
                            unpack=True)
```

In [ ]:
```
# TEST : normal vectors visualization.
# plt.quiver(XX, YY, Vx, Vy)
# plt.plot(XB, YB)
# for _ in range(XB.shape[0]):
#     plt.plot([XB[_], XB[_] + 0.05*nX[_]],
#              [YB[_], YB[_] + 0.05*nY[_]], c = 'red')
# # plt.ylim(-0.75, 0.75)                    # Smaller scale makes the n
# plt.show()
```

In [ ]:
```
# TEST : choice of observation locations
# Vx[XX == min(XX)]
# nLobs = 10;
# nMobs = 2
# step = int(30/nLobs)
# plt.quiver(XX, YY, Vx, Vy)
# plt.scatter(XX[XX==min(XX)][::step], YY[XX==min(XX)][::step],
#             s = 10, c = 'red', marker = 'x')
# plt.scatter(XX[XX==max(XX)][::step], YY[XX==max(XX)][::step],
#             s = 10, c = 'red', marker = 'x')
# plt.scatter(XX[YY==min(YY)][int(30/nMobs)::int(30/nMobs)], YY[YY==min(Y
#             s = 10, c = 'red', marker = 'x')
# plt.scatter(XX[YY==max(YY)][int(30/nMobs)::int(30/nMobs)], YY[YY==max(Y
#             s = 10, c = 'red', marker = 'x')
# plt.fill(XB, YB)
```

In [ ]:
```
# TEST : obtain continuity collocation points
# nc = 10;                                                  # Discretization
# airfoilshape = Polygon(np.column_stack([XB, YB]))         # Defining a pol
# xgrid = np.linspace(min(XX)-0.1,max(XX)+0.1,nc)
# ygrid = np.linspace(min(YY)-0.1,max(YY)+0.1,nc)
# xgrid, ygrid = np.meshgrid(xgrid, ygrid)                  # raw grid of po
# airfoilcontains = lambda x, y: airfoilshape.contains(Point(x, y))   # F
# withinairfoil = np.array(list(map(airfoilcontains,
#                                   xgrid.flatten(),
#                                   ygrid.flatten()))))
# Cx = xgrid.flatten()[~withinairfoil]       # retaining only the points
# Cy = ygrid.flatten()[~withinairfoil]       # retaining only the points
# plt.scatter(Cx, Cy, marker = 'x', s = 10)
# plt.plot(XB, YB)
# plt.show()
```

Defining relevant points in the extended design space.

- Observation points (`Obs`). $v_x \rightarrow$ `m = (1,0)` . $v_y \rightarrow$ `m = (0,1)` .
- Continuity collocation points (`C`). $\phi_{xx} \rightarrow$ `m = (2,0)` . $\phi_{yy} \rightarrow$ `m = (0,2)`
- Boundary collocation points (`B`). `m = (1,0)` & `m = (0,1)` .
- Prediction points (`Pred`). `m = (1,0)` & `m = (0,1)` .

In [68]:
```
# USER INPUTS
## Observation locations with derivative orders.
nobs = 20                                       # number of observations (at i
nm = 20                                         # number of observations in th
nc = 5                                          # discretization for continuit
nb = 30                                         # discretization for boundary
UQFLAG = 1                                      # For uncertainty prediction
ostep = int(np.sqrt(XX.shape[0])/nobs)          # step size to select observat
mstep = int(np.sqrt(XX.shape[0])/nm)            # step size to select observat
bstep = int(XB.shape[0]/nb)                     # step size to select boundary
```

In [69]:
```
# Observation locations
mask = np.zeros_like(XX, dtype = bool)      # mask to choose the right ob

for _ in np.where(XX==min(XX))[0][::ostep]:
```

```
    mask[_] = True;                          # observations at -infinity.
for _ in np.where(XX==max(XX))[0][::ostep]:
    mask[_] = True;                          # observations at +infinity.
for _ in np.where(YY==min(YY))[0][mstep::mstep]:
    mask[_] = True;                          # (upper) mid-domain observat
for _ in np.where(YY==max(YY))[0][mstep::mstep]:
    mask[_] = True;                          # (lower) mid-domain observat

nobs = np.sum(mask)                          # redefining nobs based on th
Obs = np.zeros((2*nobs, 2, 2))               # Observations of [Vx, Vy].

## Setting the appropriate locations.
Obs[:nobs, 0, 0] = XX[mask];
Obs[:nobs, 0, 1] = YY[mask]
Obs[nobs:, 0, :] = Obs[:nobs, 0, :]          # repeating the same observat

## Setting the right order of derivatives.
Obs[:nobs, 1, 0] = 1; Obs[nobs:, 1, 1] = 1  # m = (1,0) and m = (0,1)
```

@David's experiment. Random observations instead of a uniform grid.

Conclusion: Doesn't give that good results. The effect of few observations is that there is a lot of domain entry-exit behaviour (stochastic). Random or uniform grid barely makes a difference in that context.

In [ ]:
```
# To use random observations at infinity instead of a uniform gird of obs
# Observation locations
# mask = np.zeros_like(XX, dtype = bool)      # mask to choose the right

# for _ in np.random.choice(np.where(XX==min(XX))[0], nobs):
#     mask[_] = True;                         # observations at -infinity
# for _ in np.random.choice(np.where(XX==max(XX))[0], nobs):
#     mask[_] = True;                         # observations at +infinity
# for _ in np.random.choice(np.where(YY==min(YY))[0], nm):
#     mask[_] = True;                         # (upper) mid-domain observ
# for _ in np.random.choice(np.where(YY==max(YY))[0], nm):
#     mask[_] = True;                         # (lower) mid-domain observ

# nobs = np.sum(mask)                         # redefining nobs based on
# Obs = np.zeros((2*nobs, 2, 2))              # Observations of [Vx, Vy].

# ## Setting the appropriate locations.
# Obs[:nobs, 0, 0] = XX[mask];
# Obs[:nobs, 0, 1] = YY[mask]
# Obs[nobs:, 0, :] = Obs[:nobs, 0, :]         # repeating the same observ

# ## Setting the right order of derivatives.
# Obs[:nobs, 1, 0] = 1; Obs[nobs:, 1, 1] = 1  # m = (1,0) and m = (0,1)
```

In [ ]:
```
# TEST : Is the `mask` vector properly defined?
# plt.scatter(XX[mask], YY[mask], marker = 'x')
# plt.quiver(XX[mask], YY[mask],
#            Vx[mask], Vy[mask])
# plt.show()
```

when choosing continuity collocation points, set the discretization based on the scale of the x-axis and y-axis. The discretization in our implemention differs by a factor of four

which we found as,

```
{python}
# setting the discretization to be representative of the
scale on x-axis and y-axis
nc = 5
# A particular constant. Discretization on both axes is a
multiple of `nc`.
rxy = (max(XX) - min(XX))/(max(YY) - min(YY))
# ratio of the scale on x-axis and y-axis.
nX = rxy * nc
# WLOG, assuming the x-scale is higher.
nY = nc
```

In [70]:
```
# continuity collocation points
airfoilshape = Polygon(np.column_stack([XB, YB]))   # Defining a polygon
# discretization on x and y axis is chosen as nY/nX = range(Y)/range(X)
xgrid = np.linspace(min(XX),max(XX),4*nc)           # collocation coordin
ygrid = np.linspace(min(YY),max(YY),nc)
xgrid, ygrid = np.meshgrid(xgrid, ygrid)            # raw grid of points.
airfoilcontains = lambda x, y: airfoilshape.contains(Point(x, y))   # Fun
withinairfoil = np.array(list(map(airfoilcontains,
                                  xgrid.flatten(),
                                  ygrid.flatten())))
Cx = xgrid.flatten()[~withinairfoil]                # retaining only the
Cy = ygrid.flatten()[~withinairfoil]                # retaining only the
p1 = Cx.shape[0]                                    # number of collocati
C = np.zeros((2*p1, 2, 2))                          # initializing the co

# Setting the locations
C[::2, 0, 0] = Cx; C[::2, 0, 1] = Cy
C[1::2, 0, :] = C[::2, 0, :]                        # locations repeating

# Setting the right order of derivatives.
C[::2, 1, 0] = 2; C[1::2, 1, 1] = 2                 # m = (2,0) & m = (0,
```

In [ ]:
```
# TEST : Are the continuity collocation points correct?
# plt.scatter(C[::2, 0, 0], C[1::2, 0, 1], marker = 'x', s = 10)
# plt.plot(XB, YB)
# plt.show()
```

In [71]:
```
# boundary collocation points
Bx = XB[::bstep]; By = YB[::bstep]                  # chosen boundary coo
p2 = Bx.shape[0]                                    # NUM of collocation
B = np.zeros((2*p2, 2, 2))                          # initialize: sparse
B[::2, 0, 0] = Bx; B[::2, 0, 1] = By
B[1::2, 0, :] = B[::2, 0, :]                        # locations repeating
B[::2, 1, 0] = 1; B[1::2, 1, 1] = 1                 # m = (1,0) & m = (0,

# normal vectors
nvects = np.column_stack([nX[::bstep], nY[::bstep]])
```

In [ ]:
```
# TEST : Are the boundary locations sensible?
# plt.scatter(B[::2, 0, 0], B[1::2, 0, 1], marker = 'x', s = 20, c = 'r')
# plt.plot(XB, YB)
# plt.show()
```

```
# TEST : Are the normal vectors summing to 1.0
# print(nvects[:,0]**2 + nvects[:,1]**2)
```

In [72]:
```
# prediction points
q = XX.shape[0]                        # NUM prediction points.
Pred = np.zeros((2*q, 2, 2))           # initialize the prediction location

# Setting the right locations.
Pred[:q, 0, :] = np.column_stack([XX, YY])
Pred[q:, 0, :] = Pred[:q, 0, :]

# Setting the right order of derivatives.
Pred[:q, 1, 0] = 1                          # m = (1,0)
Pred[q:, 1, 1] = 1                          # m = (0,1)

# creating a mask to set predictions inside the airfoil to zero.
withinairfoil = np.array(list(map(airfoilcontains,
                                  XX,
                                  YY)))
```

In [ ]:
```
# TEST : Are the prediction locations correct?
# plt.scatter(Pred[:q, 0, 0][~withinairfoil],
#             Pred[q:, 0, 1][~withinairfoil], marker = 'x')
# plt.plot(XB, YB)
# plt.show()
```

Kernel hyperparameters and covariance matrices

Set the variables `sigma` and `theta` to construct the covariance matrices OR opt for LOOCV hyperparameter tuning by setting `hypropt=1`.

In [73]:
```
# USER INPUTS
hypropt = 1;                                    # Flag to perform LOOCV h
sigma = 1.0;                                    # Default noise variance
theta = 1.0;                                    # Default lengthscale / c

# Exact observations
v1, v2 = np.zeros(p1), np.zeros(p2)             # RHS of DiffEqns
n = Obs.shape[0]                                # True velocity observati
ytrue = np.zeros(n + p1 + p2)                   # Full vector of observat
ytrue[:int(n/2)] = Vx[mask];                    # vx obs first and then v
ytrue[int(n/2):n] = Vy[mask];
ytrue[n:n+p1] = v1                              # Laplacian = 0
ytrue[n+p1:] = v2                               # normal component  = 0

# Matrix U (coefficients for both differential equations)
U1 = Umat([1, 1], p1)                           # Laplacian coefficients.
U2 = Umat(nvects, p2)                           # normal vector component

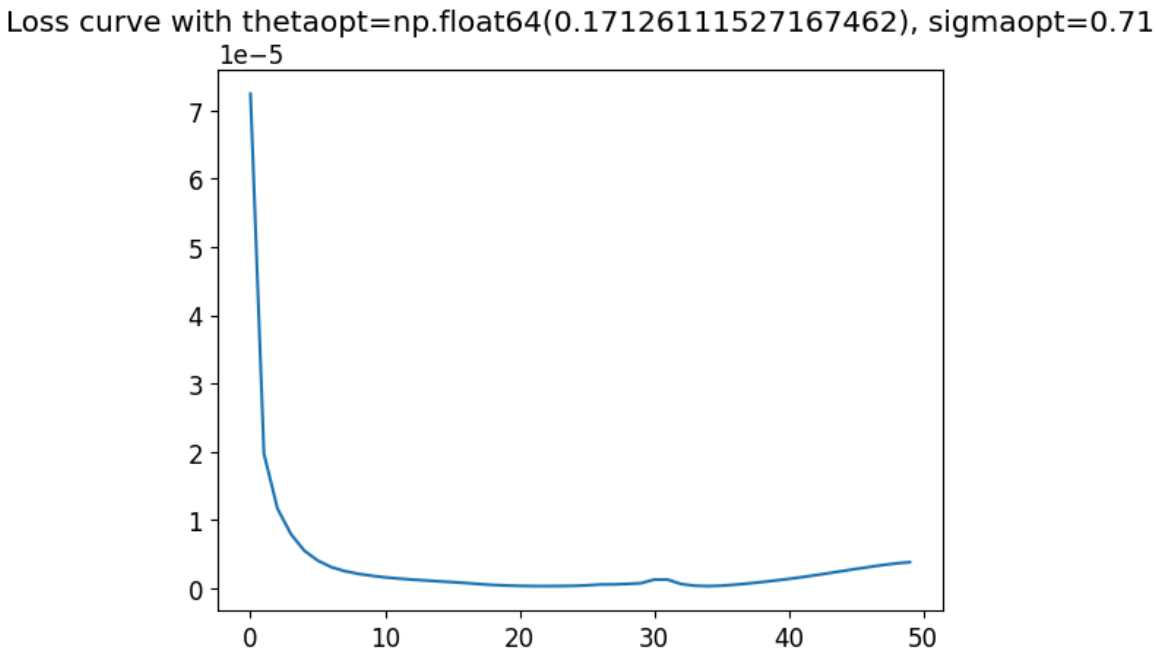
# Matrix K^+
def Kmatfn(params):
    CovParam = lambda *Z: CovM(*Z, kernel=AnalyticalSqExpdiff2D, **params
    K11 = CovParam(Obs);                        # Block matrix structure
    K12 = CovParam(Obs, C) @ U1; K21 = K12.T
    K13 = CovParam(Obs, B) @ U2; K31 = K13.T
    K22 = U1.T @ CovParam(C) @ U1
    K23 = U1.T @ CovParam(C, B) @ U2; K32 = K23.T
```

```
    K33 = U2.T @ CovParam(B) @ U2
    Kplus = np.block([[K11, K12, K13],
                      [K21, K22, K23],
                      [K31, K32, K33]])
    return Kplus
```

In [74]:

```
##########################
# LOOCV optimization step
##########################
if hypropt == 1:
    bL = 0.1; bR = 0.3                                  # left and right
    bounds = [bL, bR]                                   # To initialize t
    n_iter = 100
    loomsefn = lambda theta: LOOMSECK2D(theta, ytrue, n, Kmatfn, sigma =
    thetaopt, optscore, history = adamopt(loomsefn, bounds, n_iter = n_it
    print(f'{thetaopt=}'); print(f'{optscore=:.4f}')
    sigmaopt = LOOVARCK2D(ytrue, n, Kmatfn, dict(theta = thetaopt,
                                                 sigma = 1.0), nugget = 1e
    print(f'{sigmaopt=:.2f}')
    plt.figure()
    if not np.sum(history==1e10) == 0:                  # To check IF the
        stopindex = int(np.where(history==1e10)[0][0])  # WHERE the stopp
        plt.plot(range(stopindex - 1), history[:stopindex-1]); plt.title(
    else:
        plt.plot(range(n_iter), history); plt.title(f'Loss curve with {th
    # plt.savefig(IMGPATH + 'lossAirfoilCK.pdf')
    # plt.show()      # To visualize LOOCV loss curve
else:
    print(f'Using the fixed kernel params (user inputs): {theta=}, {sigma
    thetaopt = theta;
    sigmaopt = LOOVARCK2D(ytrue, n, Kmatfn, dict(theta = thetaopt,
                                                    sigma = 1.0), nugget
```

```
Iter 1/100 - LOOMSE: 0.000   lengthscale: 0.107
Iter 2/100 - LOOMSE: 0.000   lengthscale: 0.117
Iter 3/100 - LOOMSE: 0.000   lengthscale: 0.122
Iter 4/100 - LOOMSE: 0.000   lengthscale: 0.126
Iter 5/100 - LOOMSE: 0.000   lengthscale: 0.131
Iter 6/100 - LOOMSE: 0.000   lengthscale: 0.135
Iter 7/100 - LOOMSE: 0.000   lengthscale: 0.138
Iter 8/100 - LOOMSE: 0.000   lengthscale: 0.141
Iter 9/100 - LOOMSE: 0.000   lengthscale: 0.144
Iter 10/100 - LOOMSE: 0.000   lengthscale: 0.146
Iter 11/100 - LOOMSE: 0.000   lengthscale: 0.148
Iter 12/100 - LOOMSE: 0.000   lengthscale: 0.149
Iter 13/100 - LOOMSE: 0.000   lengthscale: 0.151
Iter 14/100 - LOOMSE: 0.000   lengthscale: 0.153
Iter 15/100 - LOOMSE: 0.000   lengthscale: 0.154
Iter 16/100 - LOOMSE: 0.000   lengthscale: 0.156
Iter 17/100 - LOOMSE: 0.000   lengthscale: 0.158
Iter 18/100 - LOOMSE: 0.000   lengthscale: 0.161
Iter 19/100 - LOOMSE: 0.000   lengthscale: 0.164
Iter 20/100 - LOOMSE: 0.000   lengthscale: 0.166
Iter 21/100 - LOOMSE: 0.000   lengthscale: 0.168
Iter 22/100 - LOOMSE: 0.000   lengthscale: 0.170
Iter 23/100 - LOOMSE: 0.000   lengthscale: 0.172
Iter 24/100 - LOOMSE: 0.000   lengthscale: 0.174
Iter 25/100 - LOOMSE: 0.000   lengthscale: 0.175
Iter 26/100 - LOOMSE: 0.000   lengthscale: 0.177
Iter 27/100 - LOOMSE: 0.000   lengthscale: 0.180
Iter 28/100 - LOOMSE: 0.000   lengthscale: 0.180
Iter 29/100 - LOOMSE: 0.000   lengthscale: 0.181
Iter 30/100 - LOOMSE: 0.000   lengthscale: 0.182
Iter 31/100 - LOOMSE: 0.000   lengthscale: 0.186
Iter 32/100 - LOOMSE: 0.000   lengthscale: 0.186
Iter 33/100 - LOOMSE: 0.000   lengthscale: 0.181
Iter 34/100 - LOOMSE: 0.000   lengthscale: 0.176
Iter 35/100 - LOOMSE: 0.000   lengthscale: 0.171
Iter 36/100 - LOOMSE: 0.000   lengthscale: 0.167
Iter 37/100 - LOOMSE: 0.000   lengthscale: 0.163
Iter 38/100 - LOOMSE: 0.000   lengthscale: 0.159
Iter 39/100 - LOOMSE: 0.000   lengthscale: 0.156
Iter 40/100 - LOOMSE: 0.000   lengthscale: 0.153
Iter 41/100 - LOOMSE: 0.000   lengthscale: 0.150
Iter 42/100 - LOOMSE: 0.000   lengthscale: 0.147
Iter 43/100 - LOOMSE: 0.000   lengthscale: 0.145
Iter 44/100 - LOOMSE: 0.000   lengthscale: 0.143
Iter 45/100 - LOOMSE: 0.000   lengthscale: 0.141
Iter 46/100 - LOOMSE: 0.000   lengthscale: 0.140
Iter 47/100 - LOOMSE: 0.000   lengthscale: 0.138
Iter 48/100 - LOOMSE: 0.000   lengthscale: 0.137
Iter 49/100 - LOOMSE: 0.000   lengthscale: 0.136
Iter 50/100 - LOOMSE: 0.000   lengthscale: 0.136
Early stopping: difference between, |iter 50 - iter 0| = 0.0001
Iter 51/100 - LOOMSE: 0.000   lengthscale: 0.135
thetaopt=np.float64(0.17126111527167462)
optscore=0.0000
sigma^2 opt. = 0.50
sigmaopt=0.71
```



Final predictions and UQ

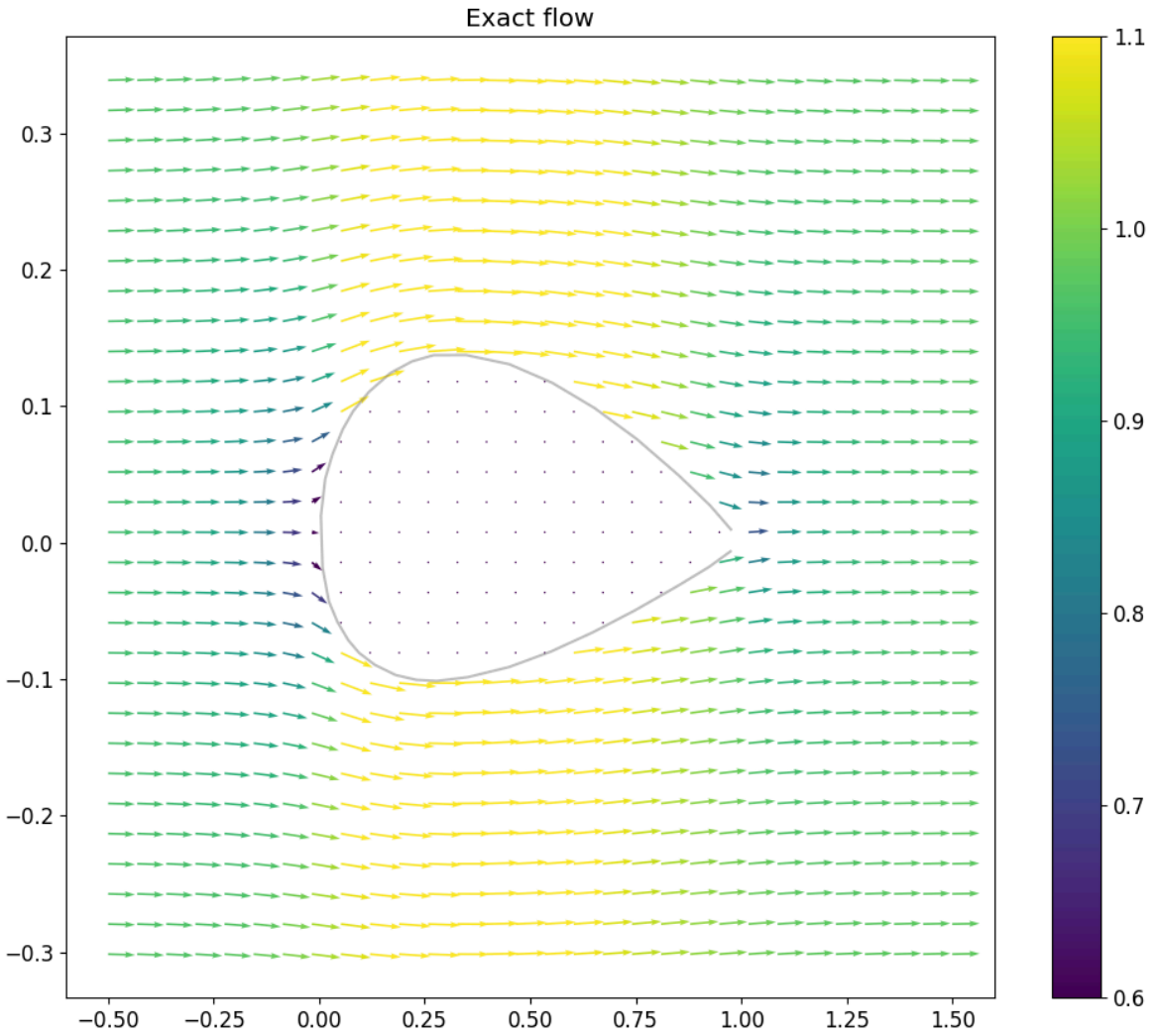
In [75]:
```
## Matrix K^+ (optimal)
Kplus = Kmatfn(dict(theta = thetaopt,                          # LOOCV Optim
                    sigma = 1.0))

## Matrix H^+
params = dict(theta = thetaopt, sigma = 1.0)
CovParam = lambda *Zvals: CovM(*Zvals, kernel = AnalyticalSqExpdiff2D, **
PredOut = np.vstack([Pred[:q][~withinairfoil],                 # vx outside
                     Pred[q:][~withinairfoil]])                # vy outside
H1 = CovParam(Obs, PredOut)
H2 = U1.T @ CovParam(C, PredOut)
H3 = U2.T @ CovParam(B, PredOut)
Hplus = np.block([[H1],
                  [H2],
                  [H3]])
```

In [81]:
```
# Simple co-Kriging [yhat]
# yhat = np.zeros(2*q)                                          # Prediction
qout = int(PredOut.shape[0]/2)
nugget = 1e-1                                                   # small nugg
I = np.eye(Kplus.shape[0]); nI = nugget * I
Kplusinv = np.linalg.solve(Kplus + nI, I)
vxhat, vyhat = (Hplus.T @ Kplusinv @ ytrue).reshape(2, qout)
# yhat = Hplus.T @ Kplusinv @ ytrue
```

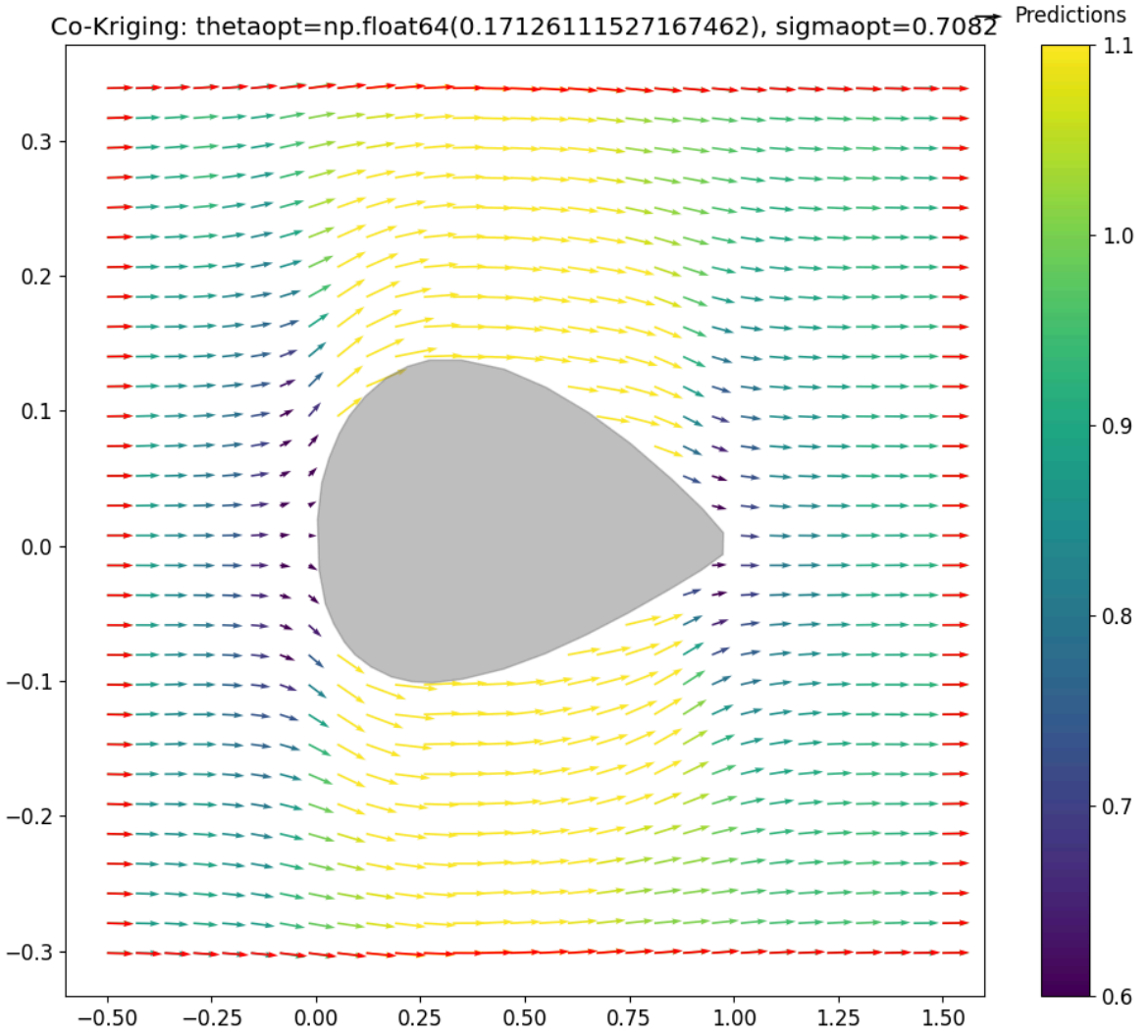
In [77]:
```
## Visualization true flow
plt.rcParams.update({'font.family': 'DejaVu Sans', 'font.size': 12})
widf = 1.5
plt.figure(figsize = (12, 10))
plt.cla()
quiverkwargs = dict(pivot = 'tail',
                    scale = 1/0.030,
                    width = 3.25/(widf*1e3),
                    clim = (0.6, 1.1))                                  #
Q1 = plt.quiver(XX, YY,
```

```
                Vx, Vy,
                np.hypot(Vx, Vy),
                alpha = 1,
                **quiverkwargs)
plt.colorbar()
plt.plot(XB, YB,
         alpha = 0.5, c = 'grey', label='Obstacle')
plt.title(f'Exact flow')
# plt.xlim(-1,2)
# plt.ylim(-1.4, 1.4)
# if figsave == 1:
#     plt.savefig(IMGPATH+'AirfoilCK.pdf', dpi = 300, bbox_inches = 'tigh
plt.show()
```



In [82]:
```
## Visualization Co-Kriging flow prediction
plt.figure(figsize = (12, 10))
plt.cla()
Q2 = plt.quiver(PredOut[:qout, 0, 0], PredOut[:qout, 0, 1],
                vxhat, vyhat,
                np.hypot(vxhat, vyhat),
                alpha = 1.0,
                **quiverkwargs)
plt.colorbar()
Q1 = plt.quiver(XX[mask], YY[mask],
                Vx[mask], Vy[mask],
                color = 'red', alpha = 1,
                **quiverkwargs)
plt.quiverkey(Q1, 1.02, 1.05, 1, label='Observations', labelpos = 'E')
```

```
plt.quiverkey(Q2, 1.02, 1.03, 1, label='Predictions', labelpos = 'E')
plt.fill(XB, YB,
         alpha = 0.5, c = 'grey', label='Obstacle')
plt.title(f'Co-Kriging: {thetaopt=}, {sigmaopt=:.4f}')
# plt.xlim(min(XX)-0.05, max(XX)+0.1)
# plt.ylim(min(YY)-0.1, max(YY)+0.1)
# if figsave == 1:
#     plt.savefig(IMGPATH+'AirfoilCK.pdf', dpi = 300, bbox_inches = 'tigh
plt.show()
```

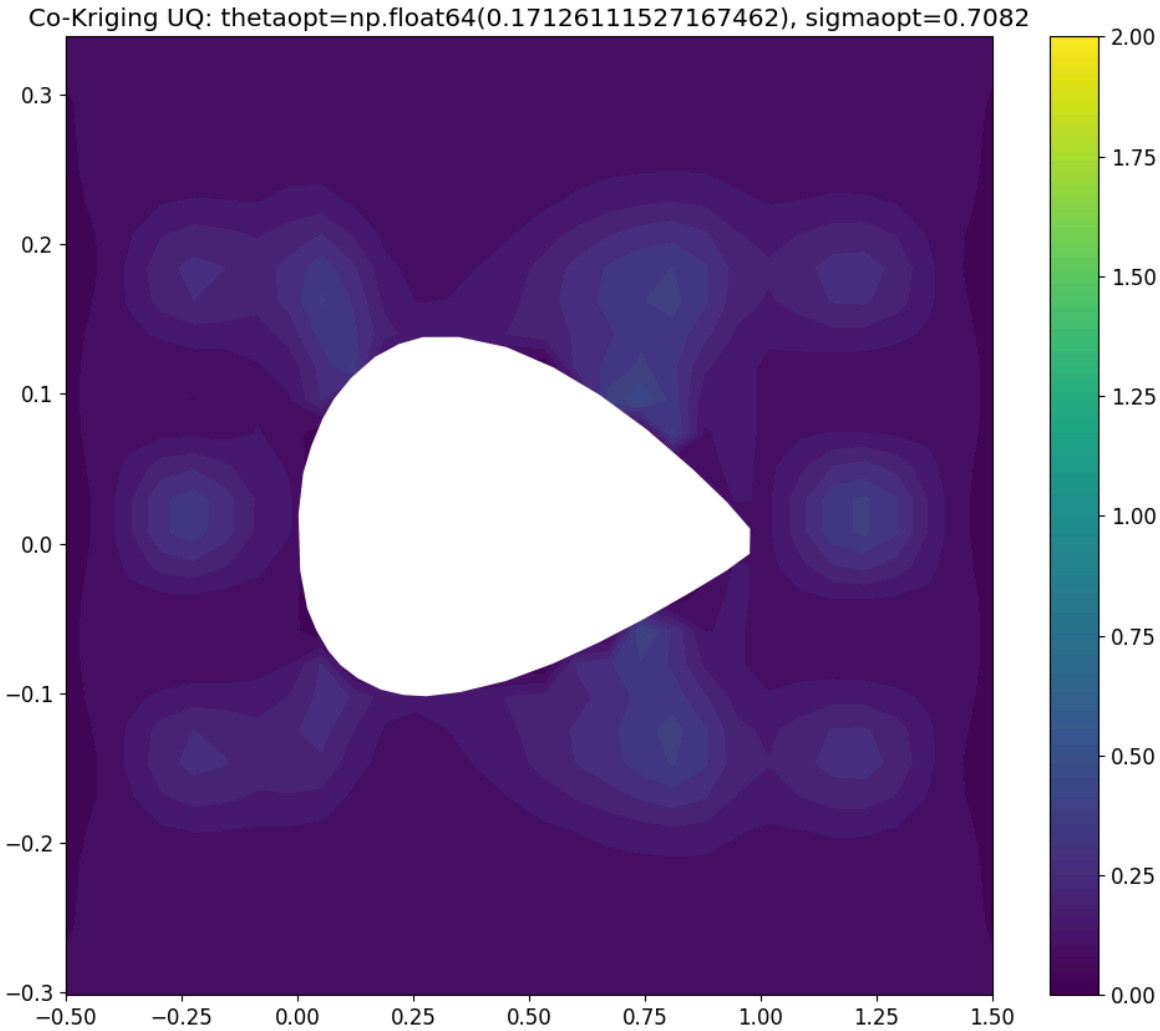


Uncertainty quantification.

In [83]:
```
if UQFLAG == 1:
    ## Uncertainty quantification: simple co-Kriging expression
    BaseVar = 1e-5                                      # To make sure C
    Kpred = CovParam(PredOut)                           # Unconditional
    CondVar = Kpred - (Hplus.T @ Kplusinv @ Hplus)      # Minimal MSE
    CondVar += BaseVar                                  # Small quantity
    CondSigma = np.sqrt(sigmaopt**2 * np.diag(CondVar));
```

In [84]:
```
## Co-Kriging uncertainty in prediction.
## Generalized chi^2 distribtion. Quadratic form of two dependent gaussia
if UQFLAG == 1:
    CondVar = sigmaopt**2 * CondVar;
    CVxVx = np.diag(CondVar[:qout, :qout])                  # Var[V
    CVxVy = np.diag(CondVar[:qout, qout:])                  # Cov[V
```

```
    CVyVy = np.diag(CondVar[qout:, qout:])                          # Var[V
    ChiVar = np.zeros(qout)                                         # Chi^2
    for i in range(qout):
        Mu = np.array([vxhat[i],                                    # predi
                       vyhat[i]])                                   # predi
        Cov = np.array([[CVxVx[i], CVxVy[i]],                       # Cov m
                        [CVxVy[i], CVyVy[i]]])
        eigvect, eigmatrix = np.linalg.eigh(Cov)
        b = eigmatrix @ Mu
        ChiVar[i] = np.sum(2 * eigvect**2 * (1 + 2*b**2))           # A der
                                                                    # A pap
    plt.figure(figsize = (12,10))
    plt.cla()
    UQ = np.zeros(XX.shape[0])                                      # initi
    UQ[~withinairfoil] = ChiVar                                     # Assig
    plt.contourf(XX.reshape(30, 30), YY.reshape(30,30),
                 2*np.sqrt(UQ).reshape(30, 30),)
                #  vmin = 0.0, vmax = 1.0)                          # Same
    cnorm = Normalize(vmin = 0.0, vmax = 2.0)                       # To no
    plt.colorbar(cm.ScalarMappable(norm=cnorm), ax = plt.gca())
    plt.clim(0.0, 2.0)                                              # Same
    # plt.colorbar()
    # plt.clim(0.0, 0.200)                                          # Same
    plt.fill(XB, YB,
            alpha = 1, c = 'white', label='Obstacle')
    plt.title(f'Co-Kriging UQ: {thetaopt=}, {sigmaopt=:.4f}')
    # if figsave == 1:
    #     plt.savefig(IMGPATH+'FluidCKUQ.pdf', dpi = 300, bbox_inches = '
    plt.show()
```


Co-Kriging UQ: thetaopt=np.float64(0.17126111527167462), sigmaopt=0.7082
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00